

MCQs on Processes and Process Management

1. What is a process?

- A) A static program
- B) An instance of a running program
- C) A type of memory allocation
- D) A hardware component
- **Answer:** B) An instance of a running program

2. Which of the following is a characteristic of preemptive processes?

- A) They run until completion without interruption.
- B) They can be forcibly interrupted by the OS.
- C) They require more memory than non-preemptive processes.
- D) They are only found in single-tasking systems.
- **Answer:** B) They can be forcibly interrupted by the OS.

3. What is the main difference between a process and a thread?

- A) Processes are lightweight, while threads are heavy.
- B) Processes share memory, while threads have their own memory space.
- C) Threads can communicate directly, while processes require IPC mechanisms.
- D) Processes are always in a running state, while threads are not.
- **Answer:** C) Threads can communicate directly, while processes require IPC mechanisms.

4. Which state indicates that a process is actively executing?

- A) New
- B) Ready
- C) Running
- D) Blocked
- **Answer:** C) Running

5. What does a short-term scheduler do?

- A) Manages the memory allocation for processes.
- B) Selects processes from the ready queue for execution on the CPU.
- C) Determines which processes are admitted to the system.
- D) Temporarily removes processes from memory.
- **Answer:** B) Selects processes from the ready queue for execution on the CPU.

6. Which scheduling algorithm executes processes in the order they arrive?

- A) Shortest Job First (SJF)
- B) Round Robin (RR)
- C) First Come, First Serve (FCFS)
- D) Priority Scheduling
- **Answer:** C) First Come, First Serve (FCFS)

7. What is a disadvantage of the Shortest Job First (SJF) scheduling algorithm?

- A) It is complex to implement.
- B) It may cause starvation of long processes.
- C) It has a high turnaround time.

- D) It does not minimize waiting time.
- **Answer:** B) It may cause starvation of long processes.

8. In Round Robin scheduling, what is the purpose of the time quantum?

- A) To determine the order of process execution.
- B) To allocate a fixed time slice for each process.
- C) To prioritize processes based on their burst time.
- D) To manage memory allocation.
- **Answer:** B) To allocate a fixed time slice for each process.

9. What is Belady's Anomaly?

- A) A situation where increasing memory leads to fewer page faults.
- B) A situation where increasing the number of page frames increases the number of page faults.
- C) A method for optimizing process scheduling.
- D) A type of process termination.
- **Answer:** B) A situation where increasing the number of page frames increases the number of page faults.

10. What does the `fork()` system call do?

- A) Terminates a process.
- B) Creates a new process by duplicating the parent process.
- C) Replaces the current process image with a new program.
- D) Waits for a specific child process to terminate.
- **Answer:** B) Creates a new process by duplicating the parent process.

11. What is a zombie process?

- A) A process that is currently running.
- B) A process that has terminated but still has an entry in the process table.
- C) A process that is waiting for I/O.
- D) A process that has been orphaned.
- **Answer:** B) A process that has terminated but still has an entry in the process table.

12. Which of the following is true about orphan processes?

- A) They are processes that have completed execution.
- B) They are adopted by the init process (PID 1).
- C) They cannot be terminated.
- D) They are always in a running state.
- **Answer:** B) They are adopted by the init process (PID 1).